

Process Report: Building Moodscape's Research Layer

Methodology and pipeline for the literature survey and the fused emotional/cultural city report

This is a methods report, not a findings report — it documents *how* the two research deliverables (the related-work survey and the fused City Mood & Character Report) were produced, so the process is auditable and repeatable, not just the output. It covers six stages: literature search, framework selection, parallel data compilation, fusion, narrative synthesis, and service integration.

Stage 1 — Literature Search

The first question, asked before any new data was collected, was whether published research already covers Moodscape's core idea (converting geographic/demographic data into music). A scoping web search across academic databases (ScienceDirect, ACM Digital Library, arXiv, PLOS One) found no paper doing exactly that, but surfaced several closely adjacent bodies of work: Chatty Maps (Aiello et al., 2016), the closest precedent for combining sound, geography, and emotion at city scale; the ISO 12913 international soundscape standard; real Korean urban-soundscape studies (Seoul, Daejeon); place-identity/city-branding research; and accessible-sonification precedents (iSonic, 2008). This survey was delivered as a standalone document before any further work began, since it determines what framing the rest of the pipeline can honestly use.

Stage 2 — Framework Selection

The brief called for the emotional data to use "a named psychology framework," with Russell's Circumplex Model suggested as a starting point. During the literature search, ISO 12913 — Russell's model formally adapted specifically for environmental/soundscape research, with Pleasantness and Eventfulness axes and eight standard descriptors — was identified as a stronger fit than generic Russell's Circumplex, since it is the instrument the actual soundscape research community uses, is directly citable, and several of the region-specific studies found in Stage 1 report their results in exactly this format. This substitution was flagged and confirmed before being used to structure the rest of the project, rather than being decided unilaterally mid-pipeline.

Stage 3 — Parallel Data Compilation

Covering all 17 Moodscape regions with real, cited data for two different axes (emotional and cultural) is a large research task. It was split three ways — six regions, six regions, five regions — and run as three parallel research passes, each instructed under an identical methodology brief:

- Search for a directly published, region-specific soundscape study first; cite it if found.

- If none exists, do not invent precise numbers presented as measured fact. Instead, produce an explicitly labeled reasoned estimate, built from real proxy data (industrial character, population density, tourism statistics, terrain), and state the reasoning.
- Compile cultural/character facts with specific numbers, dates, and named things, each cited — not generic adjectives.

The result: only 2 of 17 regions (Seoul, Daejeon) turned up a directly published soundscape study; a third (Jeju) had real tourist-perception survey data informing an otherwise estimated score. That 2-of-17 hit rate is itself a finding — it's a quantified measure of how little direct soundscape research exists for most Korean regions outside the two most heavily studied cities, and it's the reason the fused report marks 15 of 17 regions' emotional scores as estimates rather than measurements.

Stage 4 — Fusion

Per the brief's explicit instruction, the two data types were combined but kept in different formats: emotional data as a 0–100 Pleasantness/Eventfulness ratio, visualized as a circumplex scatter chart and a data table (with sourced/estimated status color- and shape-coded so the distinction survives even a quick glance at the chart), and cultural/character data as prose rather than a graph, since — as the report itself argues — cultural identity resists being reduced to a single number the way a soundscape survey response can be.

Stage 5 — Narrative Synthesis (LLM Step)

This is the step where the compiled raw facts (bullet points and citations from Stage 3) were turned into the 17 flowing paragraphs that make up Section 2 of the fused report. Each paragraph follows the same internal structure: state the region's Pleasantness/Eventfulness position in plain language and its sourcing status, then weave in 2–4 of the region's real, cited facts into a single coherent narrative rather than a bullet list. This is the literal "input the fusion into an LLM to put the data into words" step the brief asked for — done directly, not simulated, with the underlying facts and citations preserved in the report's reference list so every claim in the prose can be traced back to Stage 3's sourcing.

Stage 6 — Service Integration

The fused findings were used to rewrite the 17 region-level tagline descriptions shown in the live Moodscape app (the short subtitle under each region's name), replacing generic phrases with ones grounded in specific researched facts — e.g. Seoul's tagline now references its sourced high-eventfulness soundscape finding rather than a generic "neon-lit megacity" line. Deliberately *not* changed: the app's underlying 0–1 mood scores, which drive the map's color and the audio engine's pitch/tempo. Those remain explicitly documented (in the app's own "Research status" notice, added in an earlier pass) as hand-assigned rather than computed from either the sonification formula or this report's P/E estimates — folding a second, differently-sourced scoring system into the same numeric field the formula is supposed to eventually drive would create exactly the kind of quiet inconsistency this project has tried to avoid throughout. That integration — replacing hand-assigned scores with

formula-or-research-derived ones — remains a documented, not-yet-done next step.

Observations Worth Recording

- **Direct soundscape research on Korean cities is sparse outside Seoul and Daejeon.** This is a real finding from the process, not an assumption going in — it argues for treating Moodscape's own eventual user study (described in [about.html](#)) as a genuine contribution to a thin literature, not a redundant replication.
- **Almost every region split into at least two distinct characters** (historic/rural vs. industrial, or old-port vs. planned-city) rather than having one uniform identity — Ulsan's whaling heritage under its shipyards, Incheon's Chinatown beside Songdo, Ansan's Baekje temples near a Hyundai plant. A single P/E score per region is a real simplification of what's usually a bimodal reality; several regions' scores in this report are explicitly noted as averages of two different local profiles.
- **Running the compilation as three parallel agents rather than one sequential pass** cut wall-clock time roughly threefold for a task that scales linearly with region count, at the cost of needing a shared, explicit methodology brief up front so all three produced comparable, equally rigorous output rather than diverging in depth or standards.

What this process report is not: a claim that this pipeline constitutes a validated scientific methodology. It is a documented, repeatable procedure for compiling and fusing publicly available data under explicit honesty constraints (cite what's real, label what's estimated) — useful for a KSEF-style presentation precisely because the process itself, including its limitations, is stated plainly rather than implied to be more rigorous than it is.